Onur Altay Topaloğlu

AI / ML Engineer & Researcher · Computer Vision, Vision-Language Models & LLM Systems

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Seeking a **Werkstudent** position in Munich · **20 h/week** · No sponsorship required (student visa) · German B1 · English C1

Profile

M.Sc. Informatics student at TUM working at the intersection of computer vision research and applied AI engineering. Two years of research across three labs — monocular depth estimation, point tracking and optical flow, and fairness evaluation of vision-language models — with a CVPR 2026 workshop paper as co-first author and an AAAI 2027 submission. On the engineering side, I have shipped LLM-powered products end to end: an AI assistant with context management, prompt caching and MCP tool use, a local LLM served on Kubernetes, and hybrid RAG search. Comfortable everywhere from GPU clusters to production web stacks.

Education

10/2026 – 2028 — **Technical University of Munich (TUM)** — Munich, Germany
M.Sc. Informatics

09/2021 – 06/2026 — **Middle East Technical University (METU)** — Ankara, Türkiye
B.S. Computer Engineering (English) — CGPA **3.61/4.00**, **High Honors** (≈ 1.6 German scale)
Relevant coursework: Introduction to Deep Learning, Computer Graphics, Cloud Computing, Object-Oriented Programming
National university entrance exam: ranked 544th of 2.4M candidates (top 0.03%)

Publications

CVPR 2026 W — **LuMon: A Comprehensive Benchmark and Development Suite with Novel Datasets for Lunar Monocular Depth Estimation.**
A. Sekmen*, F. E. Güneş*, F. Höröz*, H. U. Işık*, M. A. Özaydın*, **O. A. Topaloğlu***, Ş. U. Üstündaş*, Y. A. Yeni*, H. E. Söken, E. Şahin, R. G. Cinbiş, S. Kalkan. *CVPR Workshops (AI4Space)*, 2026. *co-first authors
[Paper] [Project] [Code] [Datasets]

Under review — A construct-alignment framework for testing whether foundation-model evaluations measure the construct they claim to. Co-author (owned the fairness construct), submitted to *AAAI 2027*.

Experience

09/2025 – 07/2026 — **Harvard–MIT & METU Research Collaboration** — Remote
Undergraduate Researcher, Trustworthy Multimodal AI
Advisors: Dr. Jiaee Cheong (Harvard), Prof. Dr. Sinan Kalkan (METU) · Collaborator: Prof. Paul Liang (MIT)

- **Owned the fairness construct** of a construct-alignment framework (AAAI 2027 submission) that tests whether foundation-model evaluations measure the construct they claim to.
- Designed prompt-level demographic counterfactuals (gender, age, race / skin tone) on Harvard-FairVLMed (glaucoma, fundus images), HAM10000 (dermatoscopy) and MultiTrust; ran **7 vision-language models** (SmolVLM, Qwen2.5-VL 3B–32B, Gemma-4) with Hugging Face Transformers on A100 GPUs and via the Fireworks API.
- Built an async, resumable **LLM-as-a-judge** harness with 4 judges (GPT-4.1, Gemini 2.5 Flash, Llama-3.2-90B, Qwen-35B) and implemented group-fairness metrics (DPD, EOD, equalized odds, worst-group accuracy) and flip-rate / judge-score correlation analysis; on MultiTrust alone this meant 11K+ VLM inferences and 22K+ judge calls.
- Fairness was the most construct-aligned of 5 constructs ($r \approx -0.9$ in the non-medical setting), and **scale did not imply fairness**: Qwen2.5-VL-32B flipped 40% of glaucoma diagnoses when only the patient's age changed, vs. 3% for the 7B model.

06/2025 – 08/2025 — **KUIS AI Center, Koç University** — Istanbul, Türkiye
Research Intern, Computer Vision
Advisors: Asst. Prof. Fatma Güney

- Benchmarked three state-of-the-art **point trackers** (CoTracker3, Track-On, BootsTAPIR) on SLURM-based HPC clusters, with a shared SIFT point-seeding pipeline and side-by-side visualization for a fair comparison.
- Segmented moving animals in documentary footage with RAFT **optical flow** + FlowSAM, after showing that SAM / SAM 2 prompting was insufficient for the task.

11/2024 – 03/2026 — **ROMER — METU Center for Robotics and AI** — Ankara, Türkiye
Undergraduate Researcher, Computer Vision
Advisors: Assoc. Prof. Gökberk Cinbiş, Prof. Dr. Sinan Kalkan

- **Co-first author** of *LuMon* (CVPR 2026 AI4Space Workshop), a benchmark for lunar monocular depth estimation: 15 depth models across 6 datasets spanning real rover imagery, Earth analogs and simulators.
- Built the **stereo rectification** and masking stages of the ground-truth depth pipeline for real Chang'e-3 rover imagery: epipolar alignment from camera calibration and ORB feature matching, and masking of invalid and ambiguous pixels. The team released the resulting ground-truth depth as 2 public datasets on Hugging Face.
- Wrote the evaluation codebase: depth-vs-disparity output handling, per-image affine alignment and dataset-specific masking; integrated MiDaS v3.1 and Lotus into the benchmark.
- Drove much of the experimental analysis: metric supervision showed no significant correlation with performance ($p > 0.05$), training scale correlated significantly only on real Chang'e-3 imagery (Spearman $\rho = 0.72$), and synthetic fine-tuning gains (LuSNAR δ_1 0.31 → 0.93) did not transfer to real Chang'e-3 imagery, exposing a **sim-to-real gap**.

07/2024 **KKB Kredi Kayıt Bürosu** (Türkiye's national credit bureau) Istanbul, Türkiye
AI Engineer Intern

- Built a **PyTorch** MLP classifier predicting query–document relevance for the team's bilingual (TR/EN) internal **RAG** assistant; tuned depth, width and dropout with 5-fold cross-validation and F1-based early stopping, evaluated with F1 and ROC-AUC, and used Random Forest feature importance (MDI + permutation) to reduce its inputs from 5 to 3 features.
- Implemented BM25 keyword scoring evaluated by top-k hit rate (k = 3, 5, 10); researched hybrid search fusion (Reciprocal Rank Fusion, relative score fusion) and proposed normalized score fusion over the full corpus to replace rank averaging.
- Benchmarked SDV synthesizers (GaussianCopula, CTGAN, PAR, HMA) on public tabular data, measuring fidelity via ML efficacy for privacy-preserving financial data use.

09/2024 – 06/2025 **Computer Engineering Department, METU** Ankara, Türkiye
Undergraduate Teaching Assistant — CENG213 Data Structures (C++), CENG232 Logic Design

Engineering Projects

02/2026 – 08/2026 **LawPal — AI-native workspace for lawyers** Istanbul, Türkiye
Co-creator & Engineer, 2-person team · Next.js, React, TypeScript, Supabase, Clerk, Claude API, MCP

- Built the **AI assistant layer** on the Claude API: context injection of case metadata and extracted documents under a ∼400k-character budget, responses streamed via SSE, and **prompt caching** cutting cost and latency on follow-up turns; deferred RAG by design until real cases exceed the context budget, and enforced per-user daily AI spend caps.
- Integrated live Turkish high-court case-law search as an **agentic tool call** through an external **MCP server**, behind a user toggle that manages the trade-off with prompt caching.
- Built a **multi-tenant SaaS** workspace from scratch: workspace-based ownership with authorization verified on every server action, and soft-delete with retention instead of hard deletes.
- Built the document pipeline: private per-workspace storage and text extraction from PDF, DOCX, XLSX and the reverse-engineered UYAP court `.udf` format, with extraction failures never blocking an upload.
- Scoped the product through discovery interviews with senior lawyers of the Istanbul Bar Association; developed largely with **Claude Code**, owning the architecture and reviewing every change.

2025 – 2026 **Capture & Cook — fridge photo to recipe recommendation** METU
Senior Design Project, 4-person Scrum-style team · Flutter, Django REST, PostgreSQL, FAISS [Website]

- Compared Gemini and Qwen **vision-language model** APIs for ingredient detection on a 100-photo test set; selected Qwen for more accurate ingredient lists at similar latency and lower cost, and integrated it behind a Django REST endpoint.
- Built the **hybrid recipe search engine**: TF-IDF fused with dense semantic search over a FAISS index, fusion weights tuned on a hand-labelled human-preference set, dietary filters applied as SQL pre-filters.
- Built the Flutter mobile client (Riverpod, camera capture, Firebase Auth) and the Django REST backend with token auth enforced through reusable base view classes; wrote deploy automation for a Linux VM (systemd, git-driven) with automatic rollback on a failed deploy.

04/2026 **LLM Shopping Assistant on Kubernetes** METU
Cloud Computing coursework · Kubernetes (Minikube), Docker, Skaffold, Ollama, Python [Code]

- Deployed an 11-service polyglot microservices app (Google Online Boutique — Go, C#, Node.js, Java, Python) on **Kubernetes** with declarative manifests, orchestrated via Skaffold.
- Served a **local LLM (Ollama)** as a cluster-native service behind a new Python/Flask assistant microservice wired into the frontend; made model provisioning idempotent via a Kubernetes Job with retries and a readiness-gated init container.

03/2026 **CloudShop — cloud-deployed e-commerce web app** METU
Cloud Computing coursework · FastAPI, MongoDB Atlas, Vercel [Code]

- Built a **FastAPI** backend with JWT authentication (httpOnly cookies, bcrypt) and three server-enforced access tiers, deployed serverless on Vercel over MongoDB Atlas.

Technical Skills & Languages

Computer Vision	monocular depth estimation, stereo rectification & epipolar geometry, point tracking (CoTracker3, BootsTAPIR), optical flow (RAFT), segmentation (SAM / SAM 2, FlowSAM), vision-language models, feature matching (SIFT, ORB), diffusion & flow matching
LLMs & Agents	Claude API, MCP & tool calling, context management, prompt caching, SSE streaming, RAG & hybrid search (FAISS), LLM-as-a-judge evaluation, Ollama, Claude Code
ML & Libraries	PyTorch, Hugging Face Transformers, OpenCV, NumPy, SciPy, Pandas, Matplotlib
Programming	Python, TypeScript/JavaScript, C/C++, Java, SQL, Dart
Web & Backend	Next.js, React, FastAPI, Django REST, Flask, Flutter, Node.js
Cloud & Infra	CUDA, Slurm (HPC), Docker, Kubernetes, Skaffold, Linux/systemd, Git, PostgreSQL, MongoDB, Supabase, Vercel
Languages	Turkish (native), English (C1, TOEFL iBT), German (B1)
Interests	Calisthenics, boxing, running